CSEC PHYSICS

Jan 2022 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. Figure 1 shows the arrangement of the apparatus used to investigate how the pressure of a fixed mass of gas varies with the temperature of the gas.

   The cylinder is made of a very strong material which allows it to withstand high pressures and it is sealed so that there are no leaks. A temperature sensor is placed inside the cylinder to give an accurate record of the temperature of the gas over a wide range of temperatures. A pressure gauge is also placed in the cylinder to record the pressure readings.

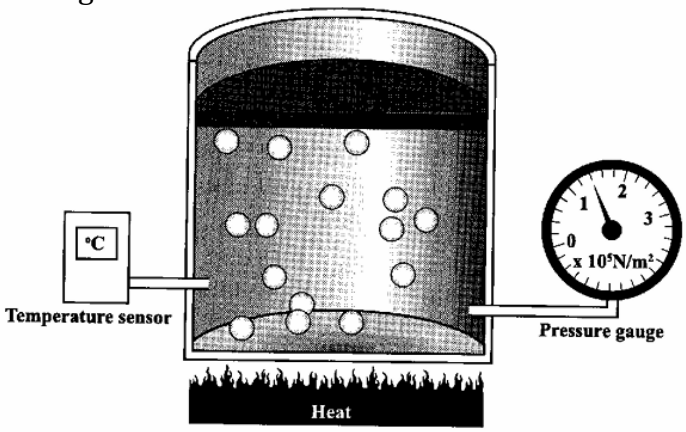

Figure 1. Apparatus used to show the relationship between the pressure and the temperature of gas

(a) Use the kinetic theory of gases to explain why the pressure of the gas increases as the temperature rises. (3 marks)

As temperature increases, the kinetic energy of the gas particles increases. The gas molecules now collide with the walls of the container more frequently and

with more velocity. This exerts a larger force on the walls over the same area.

Thus the pressure increases as temperature rises since $P = \frac{F}{A}$

Table 1 shows the results obtained when the experiment was carried out to investigate the variation of the pressure, P, (Pa) with the temperature, θ, (^{0}C) when the gas was heated.

TABLE 1: RESULTS SHOWING PRESSURE AND TEMPERATURE

Pressure, P ($\times$ $\mathbf{10^5}$ $\boldsymbol{Pa}$)	**Temperature , $\boldsymbol{\theta}$** (^{0}C)
1.2	25.0
1.3	50.0
1.4	75.0
1.5	100.0
1.6	125.0
1.7	150.0

(b) (i) Use the results from Table 1 to plot a graph of Pressure, P (Pa) against Temperature, θ, (^{0}C) on the grid provided. (6 marks)

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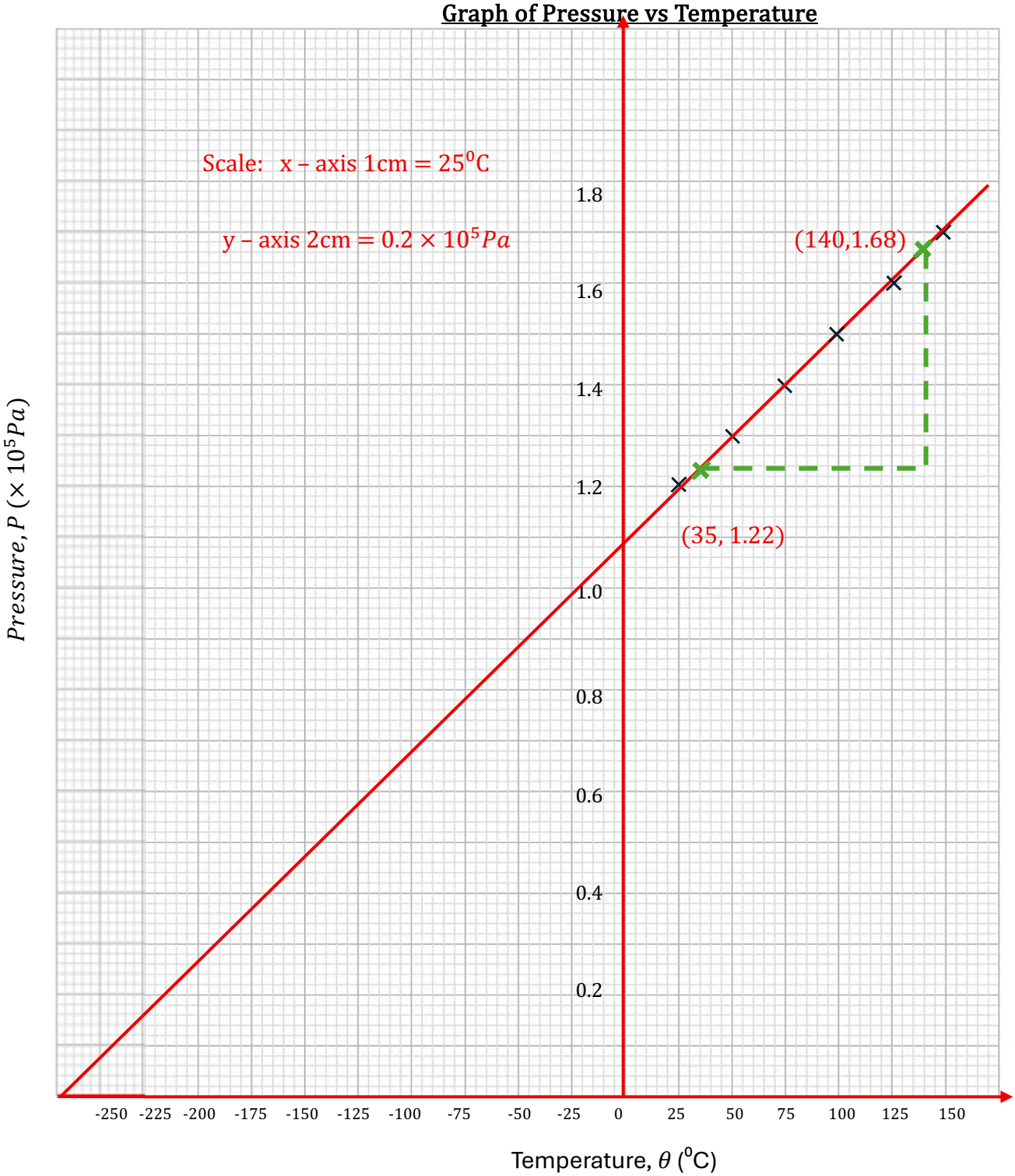
STUDENT HUB

Graph of Pressure vs Temperature



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(ii) Extrapolate the line to meet the temperature axis and mark the point X on your graph. State the temperature value.

Temperature value at point: -273^{0}C

Name of temperature value: Absolute Zero

(c) Using your graph, determine the gradient of the line (5 marks)

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

Using points (140, 1.68) and (35, 1.22) from Graph

$$Gradient = \frac{(1.68 - 1.22) \times 10^5}{140 - 35}$$

$$= 4.38 \times 10^2 \ Pa^0C^{-1}$$

(d) The sentence below Is an incomplete statement of the pressure law. Complete the statement by filling in the blank spaces.

For a _____fixed_____ mass of gas at constant _____volume_____ the _____pressure_____ is directly proportional to the _____absolute_____ temperature of the gas. (3 marks)

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(e) In the experiment on page 6, the pressure of the gas is $1.2 \times 10^5 Pa$ at a temperature of 25.0^0C. When the cylinder is heated, the pressure reaches $2.1 \times 10^5 Pa$. Calculate the temperature of the gas in ^{0}C at this pressure

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$T_2 = \frac{P_2 \times T_1}{P_1}$$

$$T_2 = \frac{(2.1 \times 10^5) \times (273 + 25)}{(1.2 \times 10^5)}$$

$$T_2 = 521.5\ K$$

$$T_2 = (521.5 - 273)^0 C$$

$$T_2 = 248.5^0 C$$

(5 marks)

Total 25 marks

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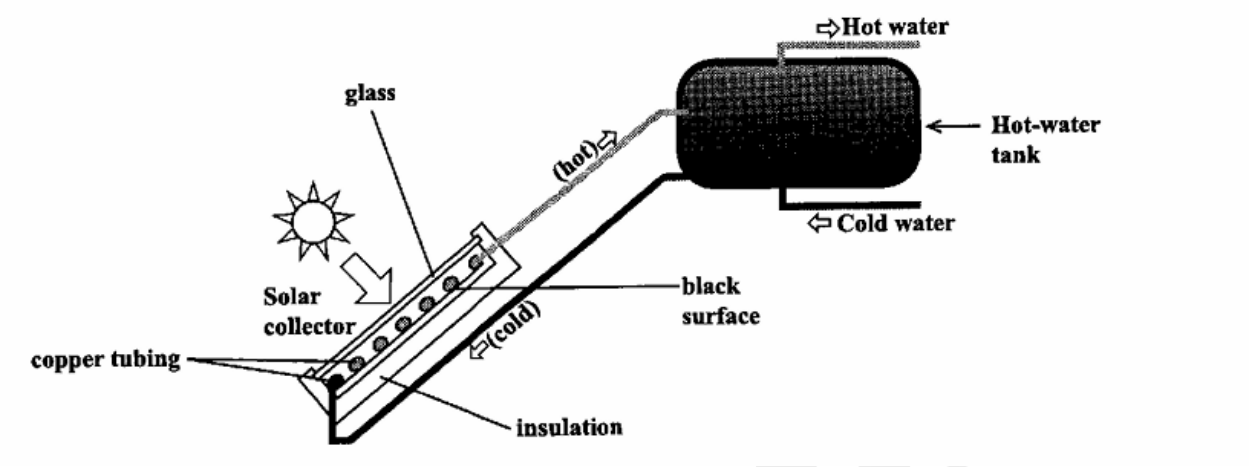
2. Figure 3 shows a sketch of a natural convection solar water heating system.



(a) State which feature of the solar water heating system demonstrates each of the following thermal energy principles (4 marks)

(i) Good absorption of heat

- a black surface

(ii) Heat transfer by convection

- Hot water tank (Hot water is less dense and remains on top)
- Solar collector – the entrapped gas transfers heat energy to the tubing

(iii) The greenhouse effect

- Glass cover – allows short wavelength radiation to enter and prevents the long wavelength radiation from leaving.

(iv) Good heat transfer by conduction

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- Copper tubing – allows the transfer of heat from the black surface absorber plate to the water in the tubing.

(b) A group of students used an electrical method to calculate the specific heat capacity of a liquid. The students found that it took 400 s for the temperature of 0.2 kg of the liquid to be raised from 25 °C to 70 ^{0}C. The energy supplied by the immersion heater was 15.6 kJ during this period.

(i) Define the term 'specific heat capacity' of a substance

The specific heat capacity, c, of a substance is the heat energy needed to raise the temperature of 1 kg of the substance by 1 Kelvin. Its equation is given by the following: $E = mc\Delta T$

(ii) Calculate the specific heat capacity of the liquid. Assume that no heat was lost and ignore the heat capacity of the container

E=15.6 kJ

Time 400s

M =0.2kg

$T_1 = 25^0C$

$T_2 = 70^0C$

Assuming no energy is lost

Energy supplied by Heater = Energy gained by the liquid.

$E = mc\Delta\theta$

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$$c = \frac{E}{m\Delta\theta}$$

$$c = \frac{15.6 \times 10^3}{(0.2)(70 - 25)}$$

$$c = 1.73 \times 10^3 Jkg^{-1}K^{-1}$$

(iii) Calculate the power rating of the immersion heater.

$$E = Pt$$

$$P = \frac{E}{t}$$

$$= \frac{15.6 \times 10^3}{400}$$

$$= 39W$$

Total 15 marks

3. (a) State the TWO laws of reflection. (3 marks)

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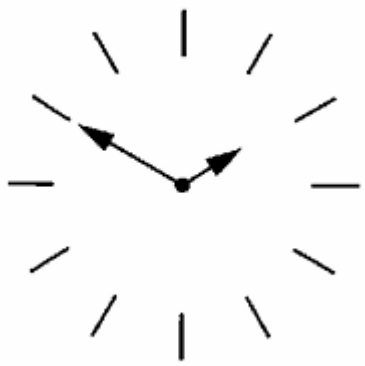
- The incident ray, the reflected ray and the normal, at the point of incidence, all lie on the same plane.
- The incident angle = The reflected angle

(b) State FOUR properties of an image formed by a plane mirror (4 marks)

- Virtual
- Image size = Object size
- Laterally Inverted
- Image distance = Object distance

(c) Maxwell washed his face at the rest room sink in his favourite restaurant. Looking into the mirror he saw the image of the clock on the wall shown in Figure 4.



State the actual time shown on the clock. (1 mark)

10:10

(d) The refractive index, n, is defined as the ratio of the speed of light in air to the speed of light in a medium, $n = \frac{V_a}{V_m}$, where m means medium and a means air.

Using this definition for the refractive index and the relationship between wavelength, frequency, and speed for a wave, derive the following formula:

$$n = \frac{\lambda_a}{\lambda_m}$$

$n = \frac{V_a}{V_m}$

$v = f\lambda$

therefore : $v_a = f\lambda_a$

$v_m = f\lambda_m$

Now

$$n = \frac{f\lambda_a}{f\lambda_m}$$

$$n = \frac{\lambda_a}{\lambda_m}$$

(e) A beam of green laser light of wavelength 532nm in air, travels in water. Calculate the wavelength of the laser light in water. (The refractive index of water is 1.33)

$n = \frac{\lambda_a}{\lambda_m}$

$\lambda_m = \frac{\lambda_a}{n}$

$\lambda_m = \frac{532 \times 10^{-9}}{1.33}$

$\lambda_m = 4 \times 10^{-7} m$

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(f) For species effects, a pool designer decides to fix the laser light at the bottom of a pool.

(i) Calculate the critical angle at the water-air boundary. (2 marks)

$$\sin c = \frac{1}{{}_w n_a}$$

$$c = \sin^{-1}\left(\frac{1}{1.33}\right)$$

$$c = 48.8°$$

(ii) If the angle of incidence of the laser light at the water – air surface is 52^0, state what happens to the light at the surface of the pool. (1 mark)

Since the angle of incidence is greater than the critical angle, total internal reflection occurs.

SECTION B

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Answer ALL questions.

4. (a) (i) Define the term 'linear momentum' and state its SI unit. (3 marks)

Linear momentum can be defined as the product of the mass of an object and its velocity. Linear momentum is a vector quantity and is given by the equation $p = mv$

SI unit is Ns or kg m s^{-1}

(ii) State the law of conservation of linear momentum (3 marks)

The law of conservation of linear momentum states that, in the absence of external forces, the total momentum of a system of bodies is constant.

Hence,

Total momentum before collision = Total momentum after collision

$$m_1v_1 + m_2v_2 = m_3v_3$$

(b) A truck, of mass 945kg, heading west on the Sir George Walter Highway slammed head on at $20ms^{-1}$ into a car of mass 630 kg, heading east.

(i) Calculate the initial momentum of the truck. (2 marks)

$$p = mv$$

$$= 945 \times (-20)$$

$$= -18\,900kgms^{-1} \quad OR\ 18\,900kgms^{-1} due\ west$$

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(ii) If both vehicles come to rest after the collision, calculate the initial velocity of the car. (4 marks)

Total momentum before collision = Total momentum after collision

$$p_1 + p_2 = p_3$$

$$m_1v_1 + m_2v_2 = m_3v_3$$

$$630v_1 - 18900 = 0$$

$$630v_1 = 18900$$

$$v_1 = \frac{18900}{630}$$

$$v_1 = 30\ ms^{-1}\ due\ east$$

(c) The velocity of the car increases from $20ms^{-1} to\ 45ms^{-1}$ in 10 s. Calculate the acceleration of the car. (3 marks)

$$a = \frac{v - u}{t}$$

$$= \frac{45 - 20}{10}$$

$$= 2.5ms^{-2}$$

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5. Figure 5 shows a solenoid that is connected to a battery with a current, I, flowing through it.

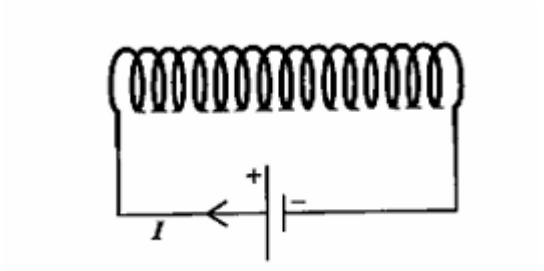

Figure 5. Solenoid connected to a battery

Figure 6 shows a cross-sectional representation of the solenoid shown in Figure 5. The symbol ⊙ represents the current coming out of the page and the symbol ⊗ represents the current going into the page

(a) Draw the magnetic field lines inside and around the solenoid on Figure 6.

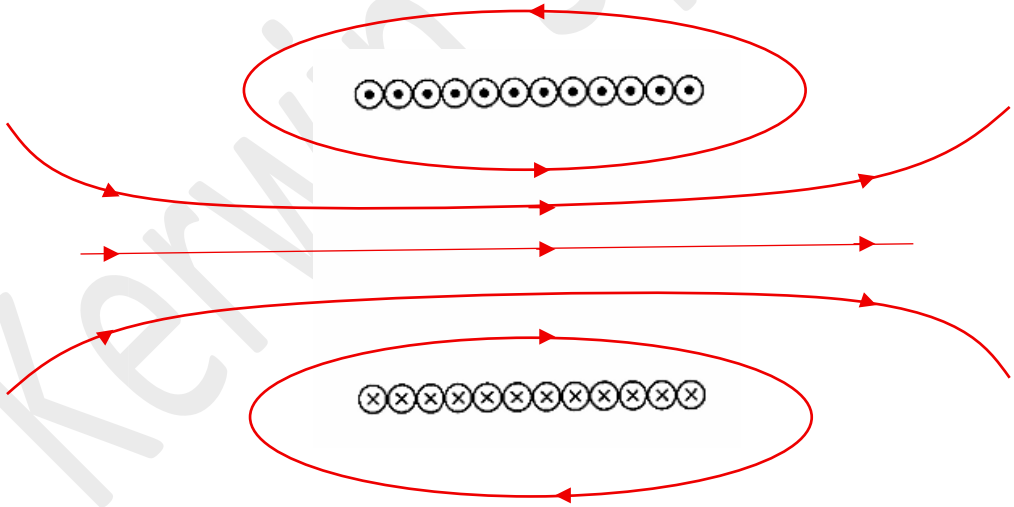

Figure 6. Cross-sectional representation of the solenoid

(3 marks)

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(b) The figures below show a constant magnetic field coming out of a surface as indicated by the symbol.

Figure 7 and Figure 8 show a straight metal wire AB moving across the magnetic field in the direction shown by the arrows, v.

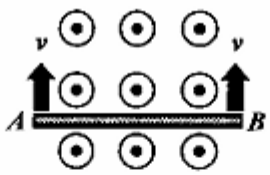

Figure 7. Wire moving across the magnetic field

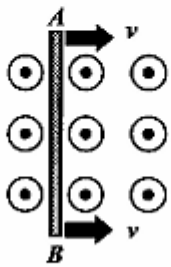

Figure 8. Wire moving across the magnetic field

Figure 9 shows the metal wire moving parallel to the magnetic field (that is, out of the surface).

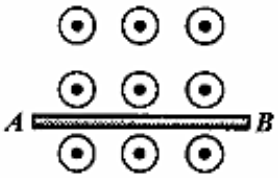

Figure 9. Wire moving parallel to the magnetic field (out of surface)

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Complete the table below by stating

(i) The direction of the induced current in the straight metal wire AB

(ii) Now the magnitude of the current would change if the speed was double

	Direction of Induced Current in the Metal Wire AB	**Magnitude of Current is Speed was Doubled**
Figure 7	From A to B (to the right)	Current doubled
Figure 8	From A to B (downwards)	Current doubled
Figure 9	No current flows	No Change

(6 marks)

(c) An ideal step down transformer is used as part of a power transmission system. It has 1980 turns in its primary coil and 180 turns in its secondary coil.

(i) Calculate the maximum theoretical output voltage if the input voltage is 1320V.

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

$$\frac{1980}{180} = \frac{1320}{V_s}$$

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$$V_s = \frac{1320 \times 180}{1980}$$

$$V_s = 120V$$

(ii) Calculate the input power if a current of 135A flows in the primary coil.

$$P_i = I_p \times V_p$$

$$= 135 \times 1320$$

$$= 178200W$$

6. (a) What is meant by the term 'nucleus of an atom? (3 marks)

The nucleus is the centre of the atom that contains the protons (positively charged) and the neutrons (no charge) of that atom.

(b) Radon- 222 is a radioactive nuclide which decays spontaneously with the emission of an alpha particle to form Polonium (Po). The nuclear equation representing the decay of a Radon - 222 nucleus is given below:

$$^{222}_{86}Rn \rightarrow {}^{A}_{Z}Po + {}^{4}_{2}\alpha$$

(i) State what is meant by the term 'spontaneously' (1 mark)

Spontaneously means that the reaction occurs on its own, without external factors.

(ii) Determine the values of A and Z (2 marks)

A:_____218 (222-4) _____

Z:_____84 (86-2) _____

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(iii) Using Einstein's mass-energy equation to calculate the energy released during this reaction. [$c = 3.0 \times 10^8 ms^{-1}$]

Masses of the nuclides: **(NOTE: Error in CSEC question, isotope Rn-228 given in table versus Rn-222 in earlier parts of question)**

Nuclide	Mass/kg
$^{228}_{86}Rn$	$368.54918 \times 10^{-27}$
$^{A}_{Z}Po$	$361.89489 \times 10^{-27}$
$^{4}_{2}\alpha$	6.64466×10^{-27}

Mass defect $= 368.841918 \times 10^{-27} - (361.89489 \times 10^{-27} + 6.64466 \times 10^{-27})$

$= 9.63 \times 10^{-30} kg$

$E = mc^2$

$= (9.63 \times 10^{-30}) \times (3.0 \times 10^8)^2$

$= 8.667 \times 10^{-13} J$

(4 marks)

(c) A fresh sample of Radon-222 gas, which has a half life of 3.8 days, was found to have a mass of 240 g.

(i) Define the term 'half life' (2 marks)

Half-life is the time taken for the number of radioactive nuclei in a sample to halve or the time taken for the number of decays, or activity to halve.

(ii) Calculate the mass of Radon-222 gas remaining after 1.52 days, giving your answer to two significant figures. (3 marks)

$$\# \; of \; half \; lives = \frac{1.52}{3.8}$$

$$= 0.4 \; half \; lives$$

$$After \; 0.4 \; half \; lives, remaining \; fraction = \left(\frac{1}{2}\right)^{0.4}$$

$$\text{Number of remaining Sample} = \left(\frac{1}{2}\right)^{0.4} \times 240g$$

$$= 181.9g$$

$$= 180g \; to \; 2 \; sf$$

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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